

Consider the following axioms:

1. All hounds howl at night.
2. Anyone who has any cats will not have any mice.
3. Light sleepers do not have anything which howls at night.
4. John has either a cat or a hound.
5. (Conclusion) If John is a light sleeper, then John does not have any mice.

The conclusion can be proved using Resolution as shown below. The first step is to write each axiom as a well-formed formula in first-order predicate calculus. The clauses written for the above axioms are shown below, using $LS(x)$ for 'light sleeper'.

1. $\forall x (HOUND(x) \rightarrow HOWL(x))$
2. $\forall x \forall y (HAVE(x,y) \wedge CAT(y) \rightarrow \neg \exists z (HAVE(x,z) \wedge MOUSE(z)))$
3. $\forall x (LS(x) \rightarrow \neg \exists y (HAVE(x,y) \wedge HOWL(y)))$
4. $\exists x (HAVE(John,x) \wedge (CAT(x) \vee HOUND(x)))$
5. $LS(John) \rightarrow \neg \exists z (HAVE(John,z) \wedge MOUSE(z))$

The next step is to transform each wff into Prenex Normal Form, skolemize, and rewrite as clauses in conjunctive normal form; these transformations are shown below.

1. $\forall x (HOUND(x) \rightarrow HOWL(x))$
 $\neg HOUND(x) \vee HOWL(x)$
2. $\forall x \forall y (HAVE(x,y) \wedge CAT(y) \rightarrow \neg \exists z (HAVE(x,z) \wedge MOUSE(z)))$
 $\forall x \forall y (HAVE(x,y) \wedge CAT(y) \rightarrow \forall z \neg (HAVE(x,z) \wedge MOUSE(z)))$

$$\forall x \forall y \forall z (\neg (HAVE(x,y) \wedge CAT(y)) \vee \neg (HAVE(x,z) \wedge MOUSE(z)))$$

$$\neg HAVE(x,y) \vee \neg CAT(y) \vee \neg HAVE(x,z) \vee \neg MOUSE(z)$$

$$3. \forall x (LS(x) \rightarrow \neg \exists y (HAVE(x,y) \wedge HOWL(y)))$$

$$\forall x (LS(x) \rightarrow \forall y \neg (HAVE(x,y) \wedge HOWL(y)))$$

$$\forall x \forall y (LS(x) \rightarrow \neg HAVE(x,y) \vee \neg HOWL(y))$$

$$\forall x \forall y (\neg LS(x) \vee \neg HAVE(x,y) \vee \neg HOWL(y))$$

$$\neg LS(x) \vee \neg HAVE(x,y) \vee \neg HOWL(y)$$

$$4. \exists x (HAVE(John,x) \wedge (CAT(x) \vee HOUND(x)))$$

$$HAVE(John,a) \wedge (CAT(a) \vee HOUND(a))$$

$$5. \neg [LS(John) \rightarrow \neg \exists z (HAVE(John,z) \wedge MOUSE(z))] \text{ (negated conclusion)}$$

$$\neg [\neg LS(John) \vee \neg \exists z (HAVE(John,z) \wedge MOUSE(z))]$$

$$LS(John) \wedge \exists z (HAVE(John,z) \wedge MOUSE(z))$$

$$LS(John) \wedge HAVE(John,b) \wedge MOUSE(b)$$

The set of CNF clauses for this problem is thus as follows:

$$1. \neg HOUND(x) \vee HOWL(x)$$

$$2. \neg HAVE(x,y) \vee \neg CAT(y) \vee \neg HAVE(x,z) \vee \neg MOUSE(z)$$

$$3. \neg LS(x) \vee \neg HAVE(x,y) \vee \neg HOWL(y)$$

4.

$$1. HAVE(John,a)$$

$$2. CAT(a) \vee HOUND(a)$$

5.

1. $LS(John)$
2. $HAVE(John, b)$
3. $MOUSE(b)$

Now we proceed to prove the conclusion by resolution using the above clauses. Each result clause is numbered; the numbers of its parent clauses are shown to its left.

- [1.,4.(b):] 6. $CAT(a) \vee HOWL(a)$
[2,5.(c):] 7. $\neg HAVE(x, y) \vee \neg CAT(y) \vee \neg HAVE(x, b)$
[7,5.(b):] 8. $\neg HAVE(John, y) \vee \neg CAT(y)$
[6,8:] 9. $\neg HAVE(John, a) \vee HOWL(a)$
[4.(a),9:] 10. $HOWL(a)$
[3,10:] 11. $\neg LS(x) \vee \neg HAVE(x, a)$
[4.(a),11:] 12. $\neg LS(John)$
[5.(a),12:] 13. $\square$

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2. Consider the following axioms:

1. Every child loves Santa.
2. Everyone who loves Santa loves any reindeer.
3. Rudolph is a reindeer, and Rudolph has a red nose.
4. Anything which has a red nose is weird or is a clown.
5. No reindeer is a clown.
6. Scrooge does not love anything which is weird.
7. (Conclusion) Scrooge is not a child.

Represent these axioms in predicate calculus; skolemize as necessary and convert each formula to clause form. (Note: 'has a red nose' can be a single predicate. Remember to negate the conclusion.) Prove the unsatisfiability of the set of clauses by resolution.

3. Consider the following axioms:

1. Anyone who buys carrots by the bushel owns either a rabbit or a grocery store.
2. Every dog chases some rabbit.
3. Mary buys carrots by the bushel.
4. Anyone who owns a rabbit hates anything that chases any rabbit.
5. John owns a dog.
6. Someone who hates something owned by another person will not date that person.
7. (Conclusion) If Mary does not own a grocery store, she will not date John.

Represent these clauses in predicate calculus, using only those predicates which are necessary. For example, you need not represent 'person', and phrases such as 'who buys carrots by the bushel' may be represented by a single predicate. Negate the conclusion and convert to clause form, skolemizing as necessary. Prove the unsatisfiability of the resulting set of clauses by resolution.

4. Consider the following axioms:

1. Every Austinite who is not conservative loves some armadillo.
2. Anyone who wears maroon-and-white shirts is an Aggie.
3. Every Aggie loves every dog.
4. Nobody who loves every dog loves any armadillo.
5. Clem is an Austinite, and Clem wears maroon-and-white shirts.
6. (Conclusion) Is there a conservative Austinite?